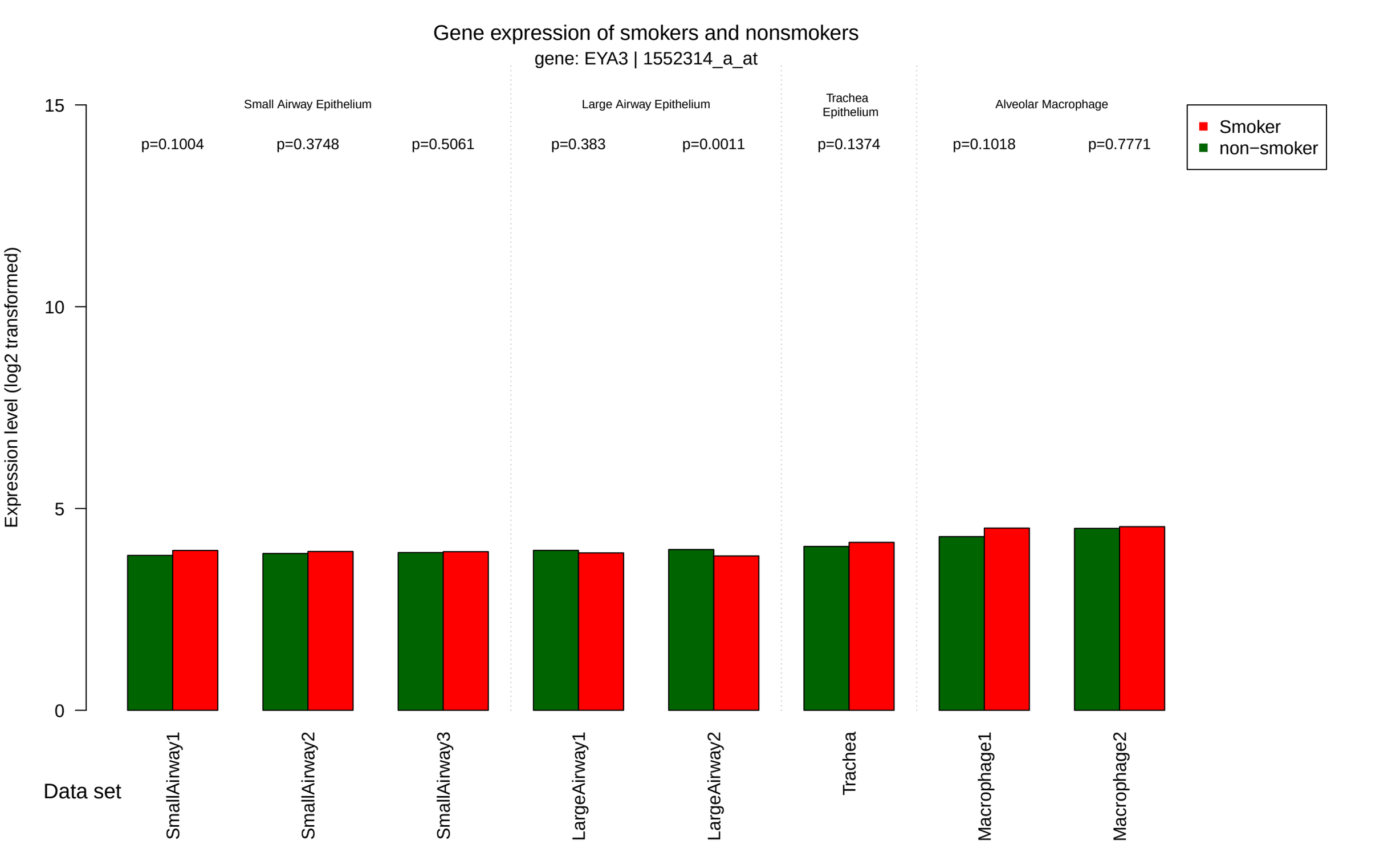
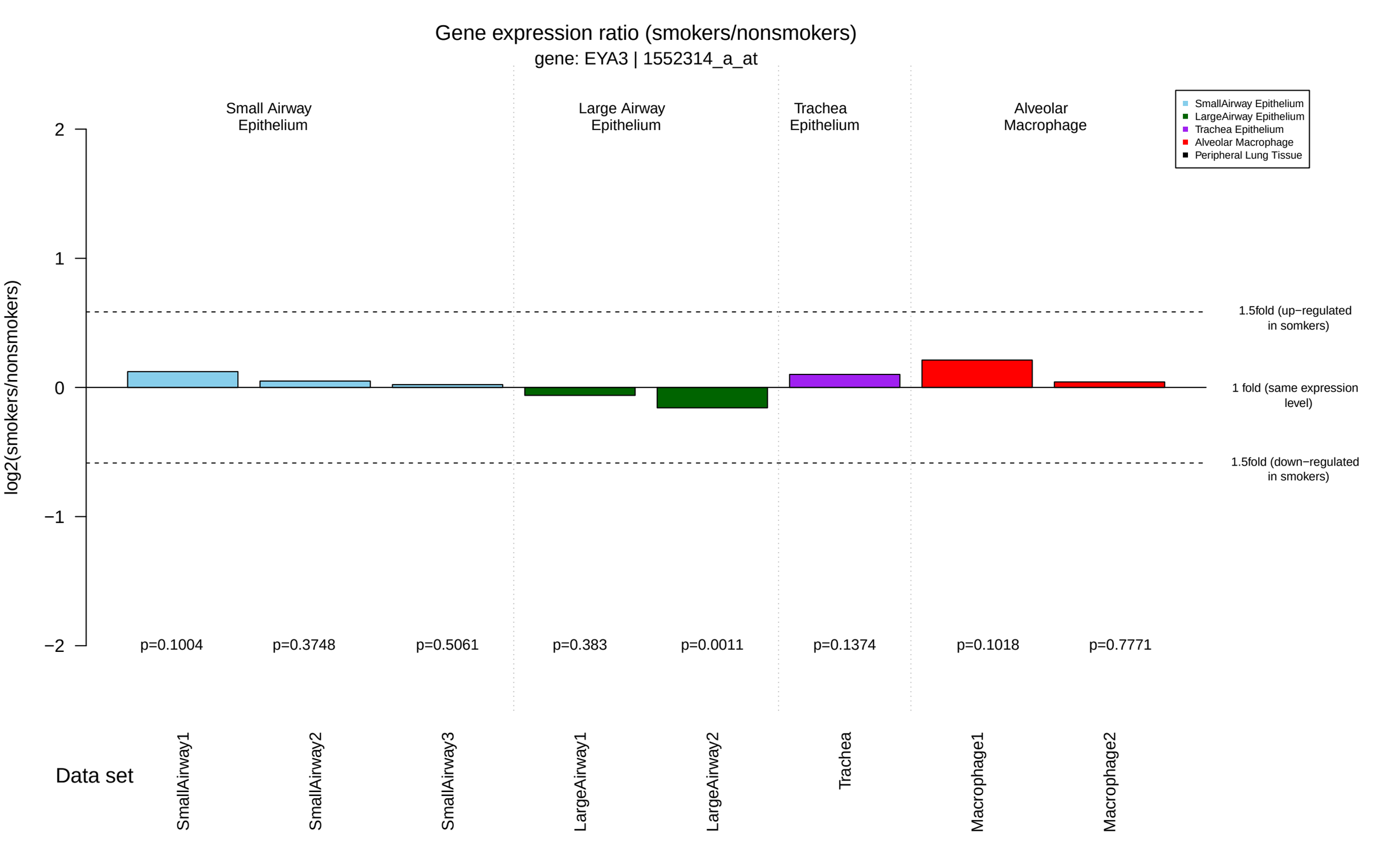
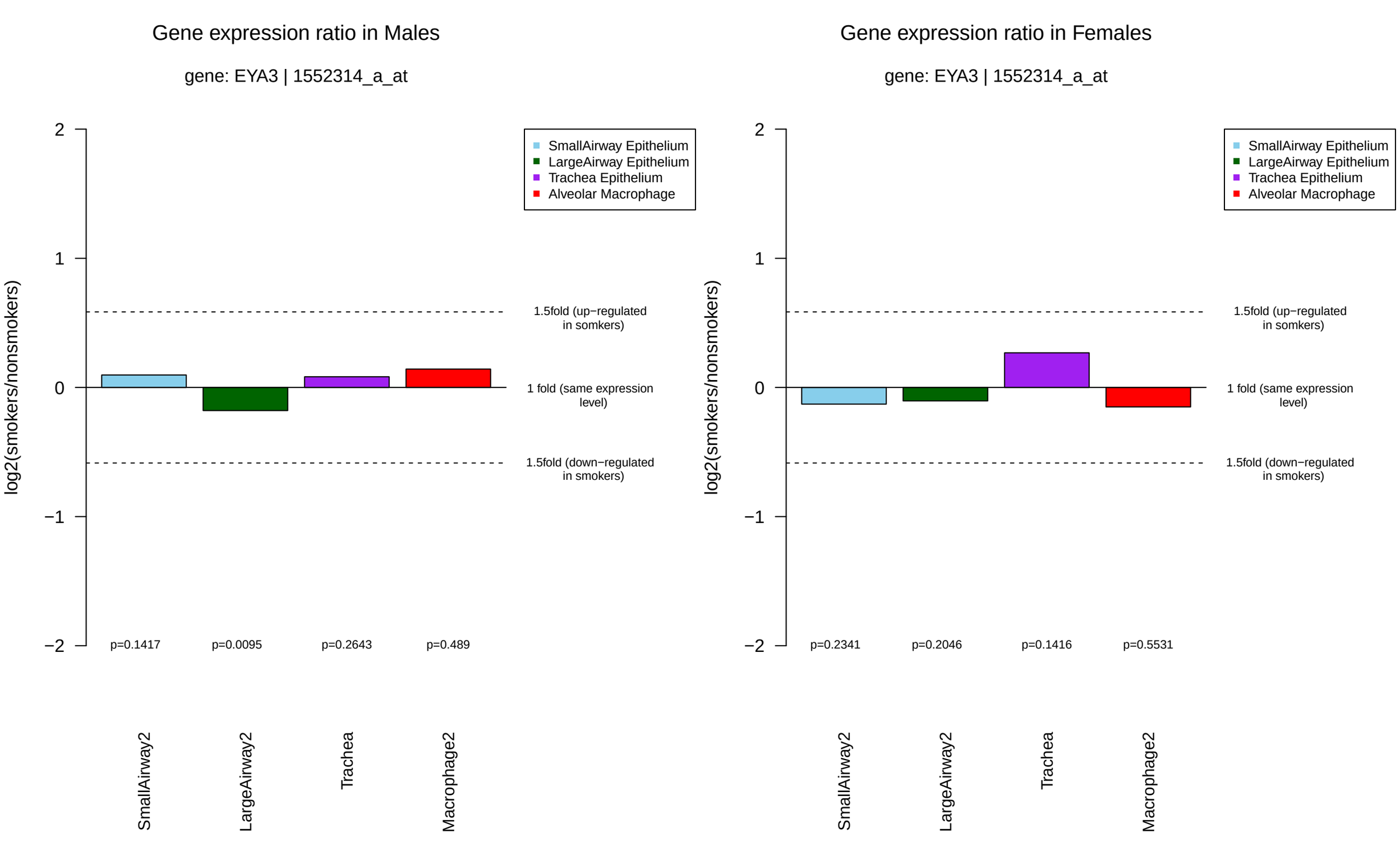
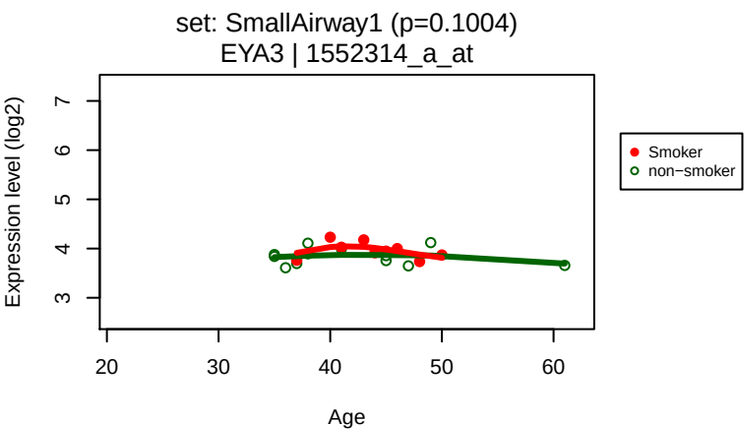
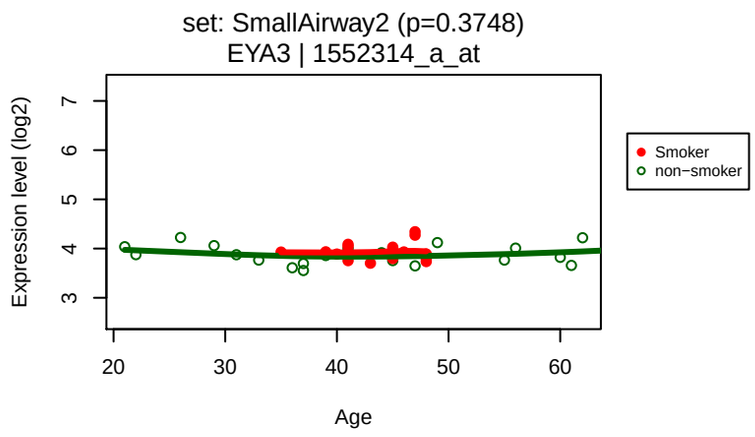
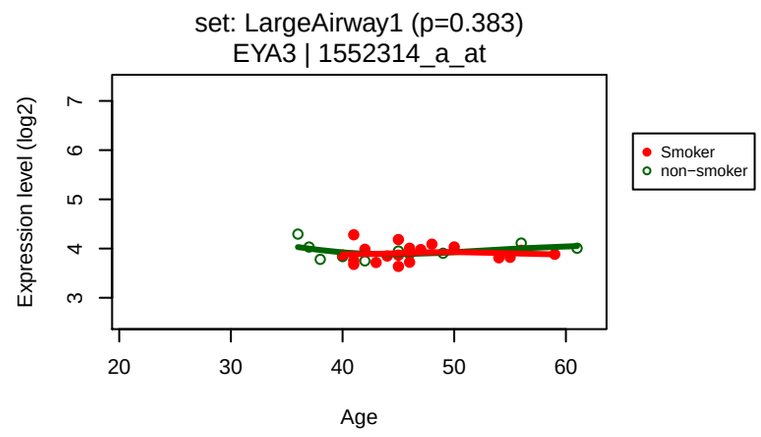

Gene expression of smokers and nonsmokers
gene: EYA3 | 1552314_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1004
p=0.3748
p=0.5061
p=0.383
p=0.0011
p=0.1374
p=0.1018
p=0.7771
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

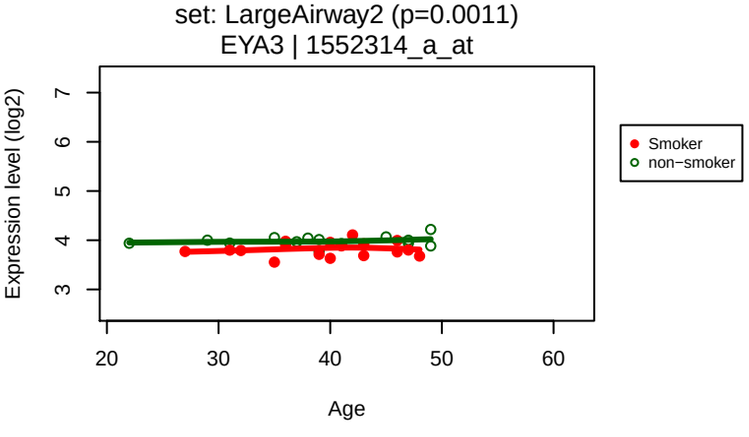

Gene expression ratio (smokers/nonsmokers)
gene: EYA3 | 1552314_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1004
p=0.3748
p=0.5061
p=0.383
p=0.0011
p=0.1374
p=0.1018
p=0.7771
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

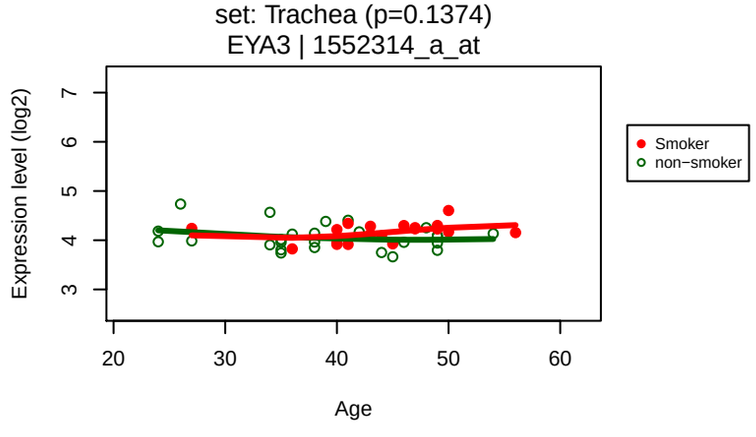

Gene expression ratio in Males
gene: EYA3 | 1552314_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1417
p=0.0095
p=0.2643
p=0.489
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EYA3 | 1552314_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2341
p=0.2046
p=0.1416
p=0.5531
SmallAirway2
LargeAirway2
Trachea
Macrophage2

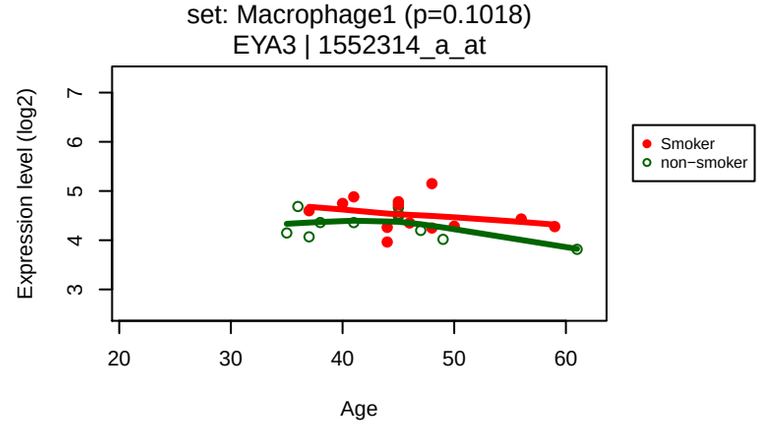

set: SmallAirway1 (p=0.1004)
EYA3 | 1552314_a_at
Expression level (log2)
Age
Smoker
non−smoker

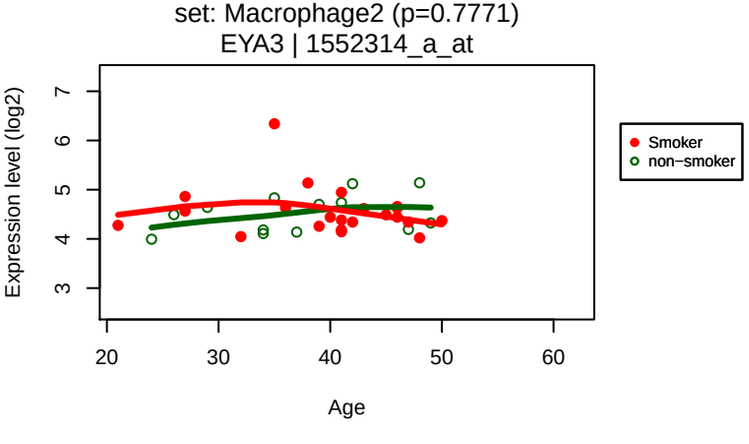

set: SmallAirway2 (p=0.3748)
EYA3 | 1552314_a_at
Expression level (log2)
Age
Smoker
non−smoker

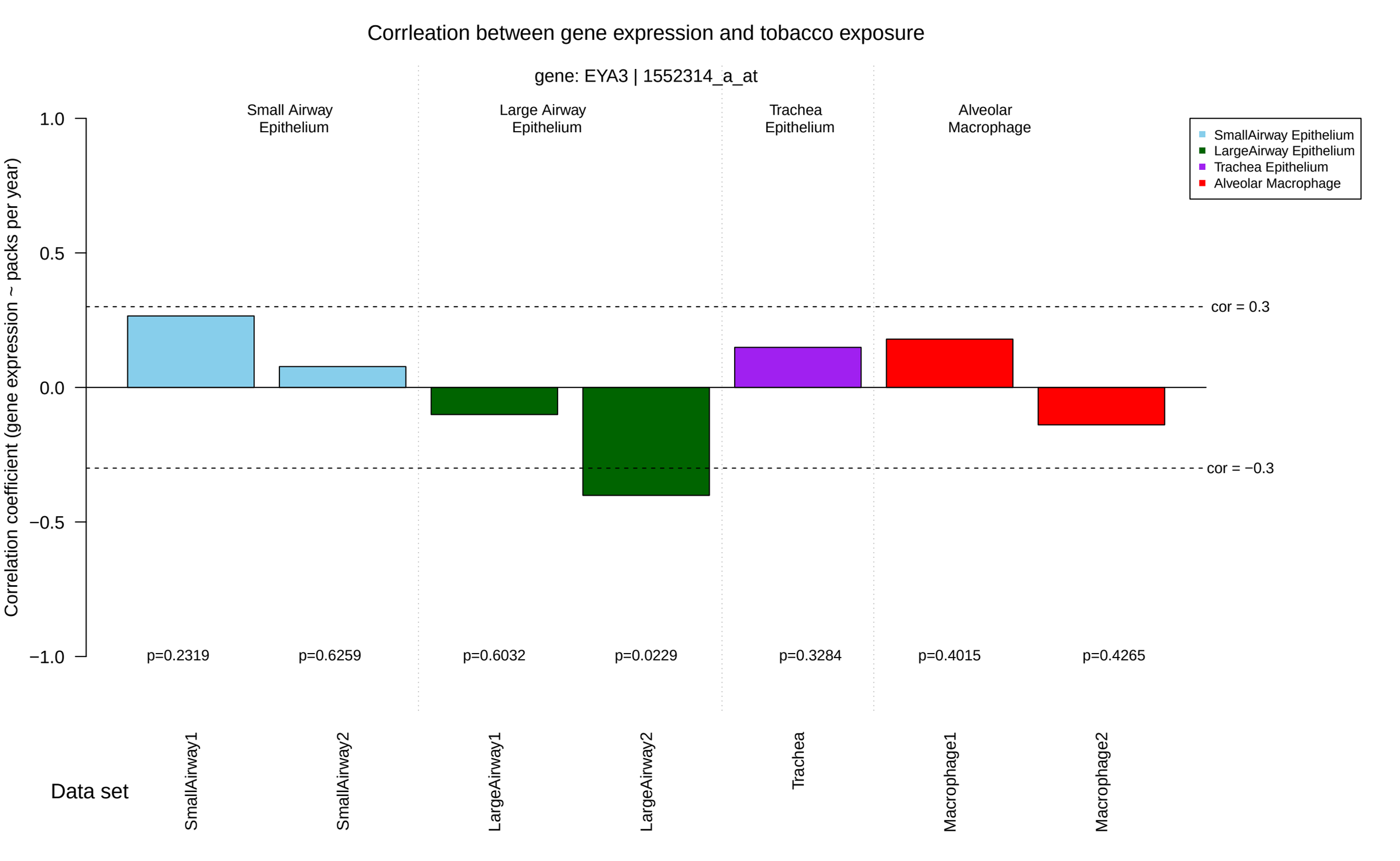

set: LargeAirway1 (p=0.383)
EYA3 | 1552314_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0011)
EYA3 | 1552314_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1374)
EYA3 | 1552314_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1018)
EYA3 | 1552314_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7771)
EYA3 | 1552314_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EYA3 | 1552314_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2319
p=0.6259
p=0.6032
p=0.0229
p=0.3284
p=0.4015
p=0.4265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2